Noise Covariance as Representation Geometry: One Metric, Four Architecture Classes

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Abstract

Label-free scientific reconstruction learns representations from measured observations alone — self-supervision in which the instrument-characterized noise covariance, not an augmentation policy, decides which residuals are informative. For Gaussian noise, inverse covariance is the likelihood-consistent reconstruction metric. Every ingredient is classical — generalized least squares, the matched filter, weighted PCA, the linear-autoencoder/PCA correspondence, multilinear PCA — and we introduce no new estimator; we organize them as a pair $(\Sigma^{-1}, \mathcal{F})$, the likelihood fixing the residual geometry and the architecture the admissible representation class. Holding the metric fixed, we walk $\mathcal{F}$ up four rungs: optimal filtering, unrestricted rank-k linear projection, Kronecker-separable two-factor projection, and factored attention. At the linear rungs the metric question has an exact geometric answer and reverses in practice: models sharing an architecture and differing only in the training metric rank oppositely under raw MSE and the likelihood residual, in a frozen ten-seed experiment and again in an independent twenty-seed replication, with a white-noise identity check at exactly zero. The covariance-weighted factors also recover the planted channel subspace almost exactly while temporal recovery stays weak. At the factored-attention rung, across two independent training seeds, the reversal does not appear, and the obvious explanation — a near-degenerate whitening there — is ruled out by measurement. The metric fixes the geometry; whether that choice bites is a property of the instrument and the representation.

1 Introduction

Many scientific instruments produce abundant measurements but no labels or clean targets: each observation is its own training signal, and an instrument-derived noise model says which discrepancies matter. Uncertainty varies across frequencies, sensors and conditions; mean-squared error (MSE) weighs all residual coordinates equally, yet under colored noise a large residual can lie in a poorly measured direction while a small one in a low-noise direction is decisive. The measurement covariance is therefore a measurement-model prior fixing the geometry of reconstruction error, distinct from a conservation law or equivariance [Bronstein et al., 2021, Karniadakis et al., 2021].

This complements self-supervised reconstruction without clean targets [Batson and Royer, 2019, Yaman et al., 2020, Gruber et al., 2025]; unlike augmentation- or masking-based self-supervision [Chen et al., 2020, He et al., 2022], which presupposes simulable nuisance structure, measurement noise supplies it directly.

Every individual link is classical; we state the boundary plainly. Inverse-covariance weighting is generalized least squares [Aitken, 1936] and, at rank one, the matched or optimal filter [Turin, 1960, Kay, 1993, Szymkowiak et al., 1993]; that literature fixes the weighting but assumes a *known*, one-dimensional template. The latent-variable line relaxes the signal model but not the noise model:

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probabilistic PCA assumes isotropic noise [Tipping and Bishop, 1999], factor analysis a diagonal covariance [Rubin and Thayer, 1982, Roweis, 1998]; weighted low-rank models [Bailey, 2012, Allen et al., 2014], Mahalanobis low-rank representation under two-way correlated noise [Zhang et al., 2019], the linear-autoencoder/PCA correspondence [Baldi and Hornik, 1989] and multilinear PCA [Lu et al., 2008] are all established. *We introduce no new estimator.* What is new is the arrangement: placing these in one Σ^{-1} metric as a ladder of admissible representation classes, the exact condition under which the Kronecker restriction is lossless (Proposition 2), and posing the same metric question to an analogous axis-factored attention network — so the question does not stop where exact analysis does.

We organize reconstruction methods as a pair $(\Sigma^{-1}, \mathcal{F})$: the measurement likelihood fixes the residual geometry Σ^{-1}, while the architecture fixes the admissible representation class $\mathcal{F}$ — including the axis along which the data are tokenized (Section 3). Holding Σ^{-1} fixed, we walk $\mathcal{F}$ up the four rungs of Section 3: optimal filtering, unrestricted rank-k linear (Proposition 1), Kronecker-separable two-factor (Proposition 2), and factored attention. The first three rungs admit exact geometric characterization (rung 3's joint optimization is nonconvex; its subproblems are exact); the fourth is examined empirically. Nonlinear representation classes beyond attention, distribution shift, and adaptive sampling remain future work.

2 Covariance-weighted representation geometry

Let $x_i = s_i + n_i \in \mathbb{R}^d$ be mean-subtracted with $n_i \sim \mathcal{N}(0, \Sigma)$, $\Sigma \succ 0$ (mean handling in Appendix B.2). For a candidate $\hat{s}_i$ the Gaussian negative log-likelihood is, up to constants,

$$\chi^2(\hat{s}_i) = (x_i - \hat{s}_i)^\top \Sigma^{-1} (x_i - \hat{s}_i). \tag{1}$$

Whitening, $\tilde{x}_i = \Sigma^{-1/2} x_i$, maps this form to $\|\tilde{x}_i - \hat{\tilde{s}}_i\|_2^2$: Euclidean projection in whitened coordinates, Σ^{-1}-orthogonal projection in measurement coordinates. We report the per-coordinate normalized residual $\bar{\chi}^2 = (Nd)^{-1} \sum_{i=1}^N \chi^2(\hat{s}_i)$, or the per-trace mean where stated.

Why label-free learning works: for independent zero-mean signal and noise, $\mathrm{Cov}(\Sigma^{-1/2} x) = \mathrm{Cov}(\Sigma^{-1/2} s) + I$ (Appendix A); adding I shifts eigenvalues, not eigenvectors, so the leading whitened eigenspace of the *noisy* data is the signal's own. In practice Σ comes from calibration or noise-only records, estimated without evaluation data; a misspecified Σ is a misspecified geometry.

Proposition 1 (Noise-weighted projection bridge)**.** *Under the model above, OF, rank-k covariance-weighted PCA (CW-PCA), and a tied noise-aware linear autoencoder minimize Eq.* (1) *over different admissible subspaces. OF fixes a rank-one template; CW-PCA learns a rank-k subspace; and the tied autoencoder parameterizes the same learned projection when its encoder is the Σ^{-1}-adjoint of its decoder.*

Proof sketch. Whitening reduces the problem to ordinary rank-k approximation, so every minimizer spans a leading eigenspace of the whitened sample covariance, and the tied Σ^{-1}-adjoint encoder follows from the same geometry. Projection onto a *fixed* rank-one template yields the OF amplitude; rank-one CW-PCA instead *learns* the leading whitened direction, and the two coincide under the separable-template and eigengap assumptions of Corollary 1. Details in Appendix A.

Metric reversal. Raw MSE replaces Σ^{-1} by I and is likelihood-correct only when $\Sigma = \sigma^2 I$; under colored noise MSE follows raw variance while Eq. (1) emphasizes high-information directions. Consequently, two equal-rank models can satisfy $\mathrm{MSE}(A) < \mathrm{MSE}(B)$ but $\chi^2(A) > \chi^2(B)$: each is optimal in a different geometry.

3 The architecture ladder

Holding the metric fixed, we walk $\mathcal{F}$ up four rungs: rungs 1–3 in general two-axis form, evidenced synthetically; rung 4 on the short-time Fourier transform (STFT) band–frame plane of a public benchmark, where Σ^{-1} is diagonal in frequency by measurement (Section 4).

Rung 1: rank-one fixed template (optimal filtering). Fixing the signal to a known template s_0, the likelihood-optimal reconstruction is the optimal filter (OF) amplitude [Gatti and Manfredi, 1986,

Turin, 1960]; the representation class is a single direction and the metric question has a closed-form answer.

Rung 2: unrestricted rank-k linear. Replacing the fixed template with a learned rank-k subspace gives covariance-weighted PCA (CW-PCA), which is also the tied noise-aware linear autoencoder: both minimize the likelihood residual over the full Grassmannian $\mathrm{Gr}(k, n_1 n_2)$ of a two-axis measurement (Proposition 1), with reconstruction $\hat{s} = PP^\top \Sigma^{-1} x$ for a Σ^{-1}-orthonormal basis P.

Rung 3: Kronecker-separable two-axis projection (NFPA). For axis-aligned noise, $\Sigma = \Sigma_2 \otimes \Sigma_1$, NFPA restricts rung 2's projection to separable factors of a two-axis measurement $Z_i \in \mathbb{R}^{n_1 \times n_2}$ — sensor channel $\times$ time in the controlled synthetic study, frequency band $\times$ time frame on the public benchmark. On the per-axis whitened observation $\tilde{Z}_i = \Sigma_1^{-1/2} Z_i \Sigma_2^{-1/2}$ it learns orthonormal bases $U_1 \in \mathbb{R}^{n_1 \times k_1}$ and $U_2 \in \mathbb{R}^{n_2 \times k_2}$, with code $C_i = U_1^\top \tilde{Z}_i U_2$ and reconstruction $\hat{\tilde{Z}}_i = U_1 U_1^\top \tilde{Z}_i U_2 U_2^\top$ mapped back through $\Sigma^{1/2}$; the joint representation is $\mathrm{range}(U_1) \otimes \mathrm{range}(U_2)$, and the admissible family is the embedded product $\mathrm{Gr}(k_1, n_1) \times \mathrm{Gr}(k_2, n_2) \hookrightarrow \mathrm{Gr}(k_1 k_2, n_1 n_2)$.

Proposition 2 (Kronecker-restricted bridge)**.** *NFPA minimizes the CW-PCA/tied-autoencoder objective restricted to Kronecker-separable two-axis subspaces. Its optimum equals the unrestricted CW-PCA optimum if and only if at least one rank-$k_1 k_2$ CW-PCA minimizer has such a separable subspace.*

Proof sketch. In whitened coordinates the factored map is the orthogonal projector onto $\mathrm{range}(U_1) \otimes \mathrm{range}(U_2)$ (vec identity, Appendix A), so NFPA is the whitened rank-$k_1 k_2$ objective restricted to that product; equality holds precisely when that set contains an unrestricted minimizer, and otherwise the gap is the cost of the architecture prior, not a failure of the metric — a cost Section 4 measures by fitting rungs 2 and 3 at matched rank. This is a covariance-aware member of the multilinear-PCA family [Lu et al., 2008], not a new factorization. Even a full-rank factor leaves the restriction active — with $k_1 = n_1$ the family is $\mathbb{R}^{n_1} \otimes \mathrm{range}(U_2)$, still separable (Appendix B.2 shows this case empirically); at $k_1 = k_2 = 1$ rung 1 is recovered under Corollary 1's assumptions (Appendix A).

Rung 4: factored attention. Rung 3's restriction on the band–frame plane reads as $\mathrm{range}(U_f) \otimes \mathrm{range}(U_m)$; rung 4's transformer performs intra-band temporal attention over frames, then cross-band attention. The attention model is an *analogous axis-factored architecture* in the same coordinates (no containment of the Kronecker family is claimed), and Σ^{-1} is diagonal in exactly the basis the first factor indexes: on this benchmark a stationary-PSD model fits the readout noise ($\chi^2/\mathrm{dof} = 1.0031$, all gates passed; Section 4). The tokenization principle: choose the token axis in which Σ^{-1} is diagonal, and the likelihood metric becomes a per-token weight rather than a dense coupling; accordingly, real and imaginary components are two features of each of 328 physical-frequency tokens, not 656 unrelated rows. The rung-4 model is a transformer [Vaswani et al., 2017] over frame patches [Nie et al., 2023], with rotary temporal embeddings [Su et al., 2024] and learned band embeddings; the metric question is posed empirically (Section 4). Injecting the optimal-filter statistic into the forward pass [Sjostrom, 2026] is compatible with this architecture but not evaluated here.

4 Evidence

NFPA under channel compression. Superseded rung-2 evidence and the TIDMAD diagnostics motivating the lift are in Appendix B.2. Rung 3 is consolidated at $C = 8 > k_c = 2$ channels, matched rank $k = k_c k_t = 6$: NFPA, an isotropically trained factored baseline of the same architecture, CW-PCA, and PCA on the same seeds. NFPA and the isotropic baseline differ only in the training metric, isolating it. It reverses: in the frozen ten-seed experiment NFPA is better under the likelihood residual (-0.0192, 95% interval $[-0.0194, -0.0190]$) and worse under raw MSE ($+0.0034$, $[+0.0034, +0.0034]$), both excluding zero. An independent twenty-seed replication of the identical configuration agrees in sign and magnitude (-0.0190, $[-0.0192, -0.0189]$; $+0.0034$, $[+0.0034, +0.0034]$); cohorts are reported separately, any pooled figure being post-hoc rather than a thirty-seed confirmatory experiment. Against unrestricted CW-PCA the residual differs by -0.0026 ($[-0.0026, -0.0025]$) — finite-sample, not evidence the Kronecker restriction is superior. The

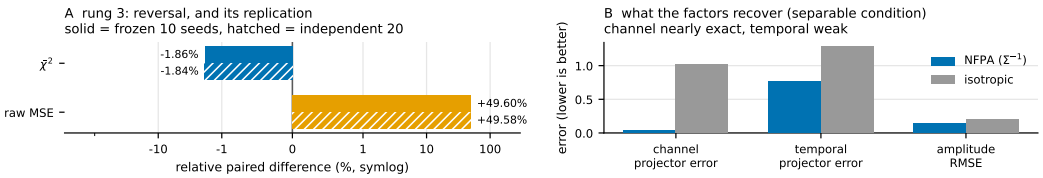


Figure 1: **A**, rung 3: the reversal and its replication. Relative paired differences, NFPA minus the isotropic baseline; solid = frozen ten seeds, hatched = an independent twenty-seed replication. The metrics point in *opposite* directions — the reversal — and the cohorts agree. Symmetric-log axis (the effects differ by over an order of magnitude); cohorts are never pooled. **B**, what the factors recover at the separable condition: the channel subspace almost exactly, the temporal one only weakly. From archived summaries via `scripts/make_rps_figure_v2.py`.

twenty-seed white-noise run is an identity check: at $\Sigma = \sigma^2 I$ the objectives coincide up to scale, and its exact zero shows the pipeline adds no spurious contrast.

Rung 4 (exploratory): the metric in a factored-attention network. One factored-attention architecture is trained on the TIDMAD readout under two objectives differing only in the metric — `T-mse` on the unwhitened spectrogram, `T-chi2` on the Σ^{-1}-weighted residual — at matched data, budget and evaluation grid, at two independent seeds. *The reversal does not appear at either seed.* The Σ^{-1}-trained arm is better on *both* criteria in both runs: held-out paired differences -1.15 and -1.17×10^{-4} on the likelihood residual, -57.2 and -57.4 on raw MSE, the latter opposite to prediction; all four signs agree across seeds. Two seeds bound run-to-run variation rather than characterise it. The release's coarse score does not separate the arms: -1.0×10^{-3}, window and file-block intervals $[-3.3, +1.4]$ and $[-4.6, +2.4] \times 10^{-3}$, both spanning zero; both sit far below the published baselines (Appendix B.1). A degenerate metric would explain it, and we tested for one: J spans a factor 441 and the whitening weights have coefficient of variation 0.69, the same order as the rung-3 geometry where the reversal *is* observed (1.10; white control 0). The metric is comparably non-degenerate at both rungs, so the absence is a property of this instantiation, not of the ladder. Capacity absorbing the weighting, optimization-trajectory effects and implicit regularization are not distinguished by two seeds.

What the covariance-weighted factors recover. The metric also changes what the representation *is*. At the separable condition NFPA recovers the planted channel subspace almost exactly (0.035 projector error against the isotropic 1.021) while temporal recovery stays weak (0.768 against 1.280) — an asymmetry a joint number would hide. All seven paired contrasts — projector errors and held-out decoding of amplitude, rise time and decay time — favour NFPA, every seed-level interval excludes zero, and all ten seeds agree in sign. No recovery threshold was predeclared, so these are continuous estimates, not a verdict (Appendix B.2).

Transfer of the noise model. On a public gravitational-wave strain release [LIGO Scientific Collaboration and Virgo Collaboration, 2016, Macleod et al., 2021], a whitened score matching its reference form to 10^{-12} and calibrated on synthetic stationary noise (dispersion 1.15) is seven to nine times over-dispersed on real H1 and L1 strain (9.42, 8.83; leave-one-out 1.77, 7.29). A covariance model can be correctly implemented, validated synthetically, and still fail to transfer: implementation identity is not calibration, and transfer is where a learned geometry is falsified.

Limitations. The ladder answers the metric question exactly at the linear rungs and empirically at rung 4, where the reversal is not observed. Rung 4 rests on two seeds per arm, bounding run-to-run variation without characterising it; both arms sit far below published full-band baselines, which are not comparable to these band-limited models; and the objectives there are measurably non-degenerate, so the absence is unexplained — capacity, optimization trajectory and implicit regularization all remain candidates. Semantic recovery is strong on the channel axis, weak on the temporal one, its decoding half post-hoc with no predeclared threshold. The linear-rung TIDMAD reversal is not replicated at a confirmatory file; stationary-PSD adequacy is per-instrument (adequate on the TIDMAD readout, $\chi^2/\mathrm{dof} = 1.0031$, failing on GW strain); and the $C = 8 > k_c = 2$ result uses an oracle covariance, bounding what a correctly specified Σ buys rather than an estimated one.

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A Proof details

A.1 Noise-weighted projection bridge

Let $W = \Sigma^{-1/2}P$. The constraint $P^\top \Sigma^{-1} P = I_k$ becomes $W^\top W = I_k$, and the reconstruction satisfies

$$\Sigma^{-1/2}\hat{s} = \Sigma^{-1/2} P P^\top \Sigma^{-1} x = W W^\top \tilde{x}.$$

The empirical loss is therefore $N^{-1}\sum_i \|\tilde{x}_i - WW^\top \tilde{x}_i\|_2^2$. By Rayleigh–Ritz, every minimizer spans a leading rank-k eigenspace of the whitened sample covariance. Mapping back yields the CW-PCA subspace. The objective depends on W only through $WW^\top$, so W is identified up to an orthogonal rotation.

A.2 Whitened signal-plus-identity decomposition

For $x = s + n$ with $s \perp n$, both zero mean, and $\mathrm{Cov}(n) = \Sigma$: independence gives $\mathrm{Cov}(x) = \mathrm{Cov}(s) + \Sigma$, and conjugating by $\Sigma^{-1/2}$,

$$\mathrm{Cov}(\Sigma^{-1/2}x) = \mathrm{Cov}(\Sigma^{-1/2}s) + I.$$

Adding I shifts every eigenvalue by one and changes no eigenvector, so the eigenvectors of the whitened data covariance are those of the whitened *signal* covariance, in the same order. Rank-k CW-PCA on noisy data therefore recovers the leading whitened signal eigenspace at the population level, which is the precise sense in which the likelihood metric supplies a self-supervised training signal.

A.3 Kronecker-restricted bridge

For every $Z \in \mathbb{R}^{C\times T}$, the vectorization identity gives

$$\mathrm{vec}(U_c U_c^\top Z U_t U_t^\top) = (U_t U_t^\top \otimes U_c U_c^\top)\mathrm{vec}(Z).$$

The Kronecker product is a symmetric idempotent matrix of rank $k_c k_t$, hence the orthogonal projector onto $\mathrm{range}(U_t)\otimes\mathrm{range}(U_c)$. NFPA is thus the ordinary whitened rank-$k_c k_t$ projection loss restricted to product subspaces. The restricted and unrestricted minima coincide exactly when an unrestricted minimizing projector belongs to this subset.

A.4 Rank-one correspondence

For $Z_i = A_i v t^\top + \epsilon_i$ with isotropic whitened noise and a unique leading signal eigendirection separated by a nonzero eigengap, the population second moment has leading vectorized direction $(t \otimes v)/(\|t\|\|v\|)$, which is feasible for rank-one NFPA. Its projection coefficient is proportional to $\langle \mathrm{vec}(Z_i), t \otimes v\rangle$, the whitened matched-filter coefficient and hence the OF amplitude up to normalization.

Corollary 1 (Rank-one correspondence)**.** *Assume isotropic whitened noise and a unique leading population signal eigendirection separated by a nonzero eigengap. If that direction is generated by a separable template $\tilde{s} = vt^\top$, population NFPA with $k_1 = k_2 = 1$ learns $v/\|v\|$ and $t/\|t\|$ up to signs; its scalar code is proportional to the OF amplitude of Section 2.*

B Experiment details and reporting limits

The controlled NFPA experiment uses $C = 2$ channels, $T = 256$ samples, a sampling rate of $1\,\mathrm{kHz}$, channel weights $(1.0, 0.65)$, and amplitude $A \sim \mathcal{N}(1, 0.3^2)$. The noise is Brownian with $S_n(f) \propto 1/f^2$ and is whitened using the matched temporal frequency-domain filter. The train and test sets contain 800 and 300 events. NFPA and IsoAE use $(k_c, k_t) = (2, 3)$, while CW-PCA and IsoPCA use rank six.

Because $C = k_c = 2$, $U_c U_c^\top = I_2$ in this experiment: the reconstruction restricts a temporal subspace shared by both channels but does not compress the channel axis. Principal angles between NFPA and CW-PCA are $0.0°$, $59.3°$, $64.6°$, $73.8°$, $84.8°$, and $87.8°$ (mean $61.7°$). The corresponding IsoAE–CW-PCA angles are $82.1°$, $83.7°$, $86.0°$, $86.6°$, $88.2°$, and $89.4°$ (mean $86.0°$). These are point estimates from one seed and split. The exact held-out raw-MSE values for IsoAE and IsoPCA are re-exported at full precision from the archived run (4.58×10^{-9} and 4.33×10^{-9}).

Table 1: Implementation check: the single-split $C = k_c = 2$ run reported by the earlier draft, retained for provenance. $\bar{\chi}^2$ is the per-coordinate normalized whitened residual $\bar{\chi}^2$ of Section 2; $\bar{\theta}$ is the mean principal angle to CW-PCA in whitened space.

Method	Training	Class	$\bar{\chi}^2$	Raw MSE	$\bar{\theta}$
CW-PCA	Σ^{-1}	full	0.9929	4.95×10^{-3}	0°
NFPA	Σ^{-1}	factored	0.9916	4.95×10^{-3}	61.7°
IsoAE	MSE	factored	0.9995	4.58×10^{-9}	86.0°
IsoPCA	MSE	full	1.0007	4.33×10^{-9}	—

Table 2: TIDMAD coarse denoising scores (higher is better) from published baselines, the archived linear estimate, and this paper's two rung-4 arms, which share architecture, data, and budget and differ only in the training objective. The two arms return bit-identical values because the upstream aggregation rounds the raw score to 0.01 before the logarithm and both band-limited arms fall in the same bin; rescoring without the rounding gives 0.0241 vs. 0.0231 in band, a contrast whose bootstrap interval spans zero (Section 4). Band-limited arms are not comparable to the full-band published rows.

Method	Coarse score
FCNet	6.55
Transformer	4.18
PUNet	3.84
none	1.10 (1.1026)
linear Wiener (archived)	-1.8025
`T-mse`	-2.7708
`T-chi2`	-2.7708

B.1 TIDMAD rung-4 baselines

A submission-strength empirical study should repeat this comparison across seeds and train/test splits and sweep SNR, noise color, channel count, rank pair, and controlled nonseparability. It should report confidence intervals for held-out $\bar{\chi}^2$, raw MSE, and subspace angles. No such uncertainty estimate is claimed here.

B.2 Predeclared confirmatory protocol

Mean handling, all experiments: the training mean is estimated once, applied unchanged to held-out observations, and added back after reconstruction.

The multi-seed confirmatory study predeclared three hypotheses. **Executed:** H1 (metric reversal at fixed architecture): under colored or correlated noise, NFPA and an isotropically trained multilinear projection with the same (k_c, k_t) should disagree in ranking, with paired across-seed confidence intervals excluding zero in the predicted directions. H2 (white-noise identity check): when $\Sigma = \sigma^2 I$ the two objectives coincide up to scale, so covariance-aware and Euclidean projectors must agree to numerical tolerance; this validates the pipeline rather than the hypothesis. **Executed later (delayed):** H3 (active restriction): with $C > k_c$ and $T > k_t$, NFPA should recover the planted channel and temporal subspaces for separable signals. This was not run with the original blocks; it was executed afterwards on the archived fits, each of which reproduces its archived held-out residual to $\leq 10^{-9}$ relative, so the analysis is computed on the archived fits rather than on refits that merely resemble them. Because the protocol set no numerical recovery threshold, the result is reported as continuous projector errors with seed-level intervals and no binary verdict. Truth-referenced parameter decoding (amplitude, rise time, decay time) was proposed in a block dropped before execution; it is reported at the archived $\alpha = 0.1$ condition as *post-hoc exploratory*, and at the separable $\alpha = 0$ condition alongside the delayed H3 analysis. An earlier design specified 30 seeds; the executed, frozen protocol is 10 seeds per the 2026-08-08 amendment, fixed before any confirmatory result existed. The primary configuration uses $C = 8$ channels, $T = 256$ samples, ranks $(k_c, k_t) = (2, 3)$, $800/400/2000$ training/validation/test events, 10 independent seeds, 10 ALS restarts per seed, SNR 10 under $\mathrm{SNR}^2 = \mathbb{E}[s^\top \Sigma^{-1} s]$, Brownian $1/f^2$ temporal noise with channel correlation, oracle covariance,

and nonseparability $\alpha = 0.1$. Results enter the main text only if every acceptance gate holds: all seeds complete without post-result configuration changes; the fixed-architecture reversal intervals exclude zero in the predicted directions; the white-noise check passes its predeclared equivalence margin; and the table and figure are generated directly from archived seed-level results.

Semantic recovery, per method. At the separable condition NFPA attains amplitude RMSE 0.143 against the isotropic baseline's 0.198 (decoded-amplitude correlation 0.70 against 0.13), and lower rise- and decay-time errors. The same analysis at the archived $\alpha = 0.1$ condition is post-hoc exploratory and agrees in direction on all seven contrasts.

The factored representation (Section 3, generic form) has $O(k_1 n_1 + k_2 n_2)$ basis parameters rather than $O(k_1 k_2 n_1 n_2)$ for an unstructured basis, trading approximation flexibility for a two-axis inductive bias related to classical tensor factorizations [Kolda and Bader, 2009, Lu et al., 2008]. Alternating spectral updates solve each factor subproblem exactly, although the joint product-manifold objective remains nonconvex.

Rung-2 numbers carried from the companion work. In the 30-seed rank-two benchmark (512 traces of dimension 256 per seed), PCA and CW-PCA had held-out weighted residuals 48.419 and 47.869 (paired difference 0.550, $[0.530, 0.571]$) while raw MSE preferred the opposite reconstruction, their whitened subspaces 86.25° apart. The ten-seed white-noise control gave a maximum angle of 1.48×10^{-6} degrees. On the TIDMAD file the amplitude RMSE fell from 0.1209 to 0.0880.